

3 Beginner AI Projects

Three hands-on AI projects that progressively introduce **LLM agents**, **document analysis**, and **computer vision** · ideal for developers starting their AI journey.

· Projects Overview

#	Project	Stack	Description
1	ReAct Agent	LangGraph + OpenAI	Tool-calling AI agent with calculator and greeting tools
2	Resume Critiquer	Streamlit + OpenAI	Upload a resume (PDF/TXT), get AI-powered feedback
3	Image Classifier	Streamlit + TensorFlow	Upload an image, get real-time MobileNetV2 predictions

· Getting Started

Prerequisites

- Python 3.10+
- OpenAI API key (for Projects 1 & 2)

Installation

```
cd 3-Beginner-AI-Projects
# Project 1
cd project1
uv sync          # or: pip install langchain langgraph
langchain-openai python-dotenv
python main.py
# Project 2
cd project2
uv sync          # or: pip install streamlit PyPDF2 openai
python-dotenv
streamlit run main.py
# Project 3
cd project3
uv sync          # or: pip install streamlit tensorflow Pillow
opencv-python numpy
streamlit run main.py
```

Environment Variables

Create a `.env` file in the project root:

```
OPENAI_API_KEY=sk-...
```

· Project Details

Project 1 · ReAct Agent (project1/main.py)

What it does: An interactive CLI agent that can call tools to answer questions.

Architecture:

User Input · LangGraph ReAct Agent · Tool Selection · Tool Execution ·
Response

```
.....  
· calculator(a, b)      ·  
· say_hello(name)      ·  
.....
```

Key Concepts:

- `@tool` decorator to define callable tools
- `create_react_agent()` from LangGraph for reasoning + acting
- ChatOpenAI (GPT-4o-mini) as the base LLM
- Interactive loop with conversation history

Logic Flow:

- 1 User enters a query
 - 2 Agent reasons about which tool to call
 - 3 Tool executes and returns result
 - 4 Agent formulates final response
 - 5 Loop continues until user types "quit"
-

Project 2 · AI Resume Critique(`project2/main.py`)

What it does: A Streamlit web app that analyzes uploaded resumes and provides targeted feedback.

Architecture:

File Upload (PDF/TXT) · Text Extraction · OpenAI GPT-4o-mini · Structured
Feedback

Streamlit UI PyPDF2 / `.read()` Role-specific prompt

Key Concepts:

- PDF text extraction with PyPDF2
- Dynamic system prompts based on target role
- Streaming OpenAI responses
- Streamlit file uploader and text display

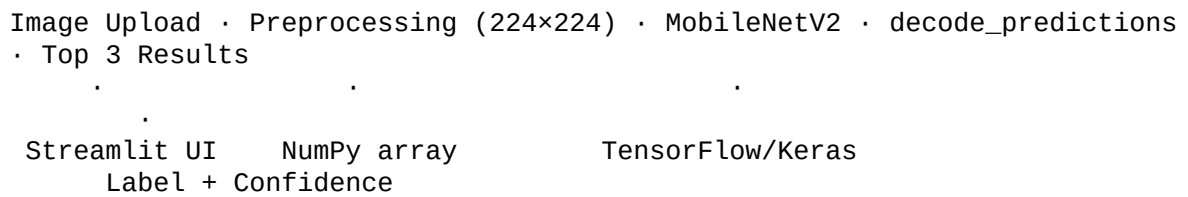
Logic Flow:

- 1 User uploads a PDF or TXT resume
 - 2 Text is extracted from the document
 - 3 User specifies the target job role
 - 4 OpenAI analyzes the resume for that role
 - 5 Feedback is displayed with formatting suggestions
-

Project 3 · Image Classifier(`project3/main.py`)

What it does: A web app that classifies uploaded images using a pre-trained MobileNetV2 neural network.

Architecture:



Key Concepts:

- Transfer learning with **MobileNetV2** (pre-trained on ImageNet)
- Image preprocessing: resize to 224x224, normalize pixel values
- `decode_predictions()` for human-readable labels
- Top-3 prediction display with confidence scores

Logic Flow:

- 1 User uploads an image (JPG, PNG)
 - 2 Image is resized and preprocessed for the model
 - 3 MobileNetV2 runs inference
 - 4 Top 3 predictions with confidence percentages are displayed
 - 5 Original image is shown alongside results
-

.. Project Structure

```
3-Beginner-AI-Projects/
... project1/
.   ... main.py           # ReAct agent with tools
.   ... pyproject.toml
... project2/
.   ... main.py           # Resume critiquer app
.   ... pyproject.toml
... project3/
.   ... main.py           # Image classifier app
.   ... pyproject.toml
```

· License

Educational project · use freely for learning and reference.